

HOMEWORK 3

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Problem 1. A complex number z is said to be algebraic if there are integers $a_0, \dots, a_n$, not all zero, such that

$$a_0 z^n + a_1 z^{n-1} + \dots + a_{n-1} z + a_n = 0.$$

Prove that the set of all algebraic numbers is countable. Hint: For every positive integer N there are only finitely many equations with

$$n + |a_0| + |a_1| + \dots + |a_n| = N.$$

Proof. To prove this, we must construct the set of all algebraic numbers, A , and create a bijection $f : A \rightarrow \mathbb{N}$. Each polynomial has at most n roots by the Fundamental Theorem of Algebra. With our hint, we may sort the infinite possible polynomials into the form of $n + |a_0| + |a_1| + \dots + |a_n| = N$, where N is some fixed value. Doing so, we can see that we get a finite set for whatever N we choose, which is the same behavior displayed with polynomials. By doing so, we can create a sequence S_N that collects all possible finite sets, and we can create a union such that $A = \bigcup_{N=1}^{\infty} S_N$, $N \in \mathbb{N}$. Since $N \in \mathbb{N}$, then there does exist a bijection $f : A \rightarrow \mathbb{N}$ $\square$

Problem 2. Prove that there exist real numbers which are not algebraic. Is the set of all irrational numbers countable?

Proof. We were told in class that $\mathbb{R}$ is uncountable. We show that $\mathbb{R} \setminus A$ is uncountable by supposing for contradiction that it is. If it were countable, then $(\mathbb{R} \setminus A) \cup A$ should be countable as well, but this results in $\mathbb{R}$, which is a contradiction. So, since $(\mathbb{R} \setminus A)$ is uncountable, there exists real numbers which are not algebraic. The set of all irrationals are not countable, by the same argument shown above, but instead of using A , we use $\mathbb{Q}$. $\square$

Problem 3. Let E' be the set of all limit points of a set E . Prove that E' is closed. Prove that E and $\bar{E}$ have the same limit points. (Recall that $\bar{E} = E \cup E'$). Do E and E' always have the same limit points?

Proof. Let p be a limit point of E' . Since p is a limit point, define its neighborhood as X . Within X , there exists another point $q \neq p$ by definition. We can fix a radius such that $q \in X$. Now, since $X \subset E'$, that means q is also a limit point. So, define Y to be a neighborhood of q , but has a smaller fixed radius than X . We do this to avoid getting the point p . Since q is a limit point, there exists a point $t \in Y$ such that $t \neq q$. If we continually do this, neighborhoods begin to cluster, alongside the points. Since we have shown for an arbitrary neighborhood X that this is true, then we may conclude that every neighborhood of p contains another point that is not itself. Thus, E' is closed.

For Part 2 of the problem, let p be a limit point in $\bar{E}$. If $p \in E$, then we're done. If $p \in E'$, then by definition, E' is the set of all limit points $\square$

Problem 4. Find a bijection between $(0, 1)$ and $[0, 1]$.

Proof. $\square$